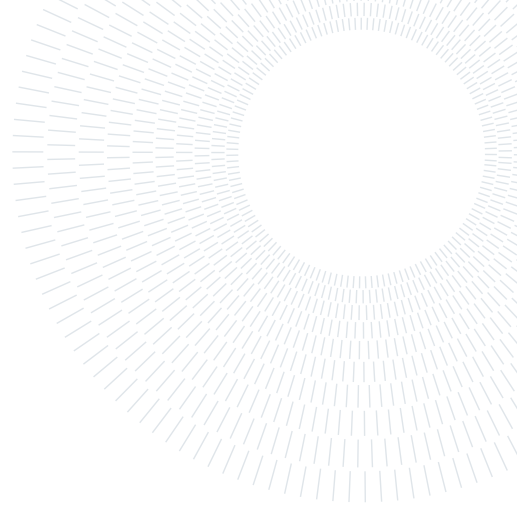




Lab Report Rotary Pendulum

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Abstract: This project addresses the modeling, control, and state estimation of an inverted pendulum, a classical benchmark in control engineering due to its inherent instability. Experimental data were obtained to identify the system parameters, which were used to derive a nonlinear model and its linearization around the upright equilibrium. Stabilization was achieved using state-feedback control via pole placement, while an optimal control strategy enabled swing-up from the downward position. A Kalman filter provided full-state estimation in the presence of measurement noise, improving control performance. Simulation and experimental results demonstrate that the proposed approach successfully achieves swing-up, stabilization, and reliable state estimation.

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1. Introduction

The inverted pendulum is a classical benchmark system in control engineering due to its inherent instability and nonlinear dynamics. Despite its simple mechanical structure, it captures many of the fundamental challenges encountered in real-world control problems, such as stabilization of unstable equilibria, controller design under modeling uncertainties, and state estimation in the presence of noise. For these reasons, it is widely used to evaluate and compare control strategies in both academic and industrial contexts.

The objective of this laboratory project is to model, control, and analyze an inverted pendulum system. The work focuses on identifying relevant system parameters from experimental data, deriving a mathematical model of the pendulum dynamics, and designing suitable control strategies to achieve both swing-up and stabilization of the pendulum in the upright position.



Figure 1: SRV02 ROTPEN System: Rotary Inverted Pendulum

To accomplish these objectives, a nonlinear model of the inverted pendulum is first derived and linearized around the upright equilibrium. State-feedback control using pole placement is then applied to stabilize the system locally. In addition, an optimal control strategy is implemented to perform the swing-up maneuver from the downward position. Where required, state estimation techniques are employed to reconstruct unmeasured states and improve control performance.

2. Theoretical Background

2.1. Inverted Pendulum System Overview

The system used in the lab is the Quanser SRV02 Rotary Pendulum. It consists of a horizontal rotary arm driven by a DC motor and an un-actuated pendulum attached to the end of that arm through a metal joint.

- Rotary arm has a length L_{arm} , a moment of inertia J_{arm} and is actuated by the servo motor Voltage V_m .
- The Pendulum has a total length of $2L_p$ a mass m_p , where L_p is the distance from the center of mass to the axis of rotation. The metal joint has its own mass m_m and a length L_m that represents the distance from its center of mass to the same axis of rotation.
- The system uses optical encoders (sensors) to measure two specific angles: the arm angle (θ) and the pendulum angle (α).

The system possesses two degrees of freedom (DOF), represented by the generalized coordinates $q = [\theta, \alpha]^T$:

- **Arm Angle (θ):** Represents the horizontal rotation of the base and **Pendulum Angle (α):** Represents the vertical rotation of the link, where $\alpha = 0$ is the perfectly upright position.

To make the mathematical modeling feasible, the following assumptions are applied:

- **Rigid Bodies:** Both the rotary arm and the pendulum are treated as rigid links.
- **Planar Motion:** The pendulum is assumed to swing within a single vertical plane relative to the arm.

- **Point Mass:** For potential energy calculations, the pendulum's mass is assumed to be concentrated at its center of gravity (L_p).
- **Linear Damping:** Friction in the system is modeled as viscous damping coefficients B_r (for the arm) and B_p (for the pendulum).

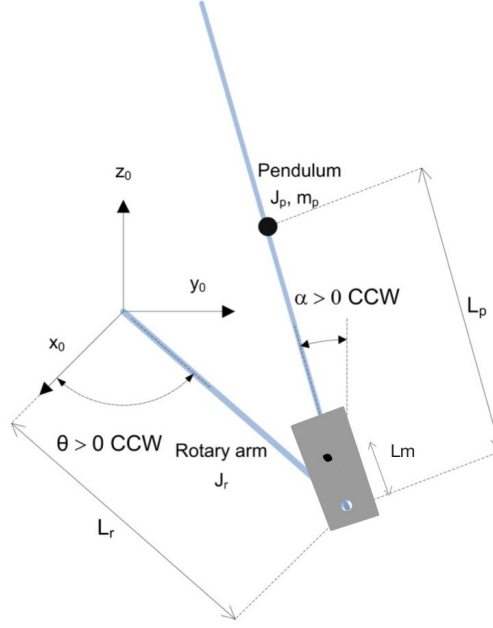


Figure 2: Rotary inverted pendulum conventions

2.2. Dynamic Modeling of the Inverted Pendulum

The dynamics of the rotary inverted pendulum are derived using the Euler-Lagrange method. This approach uses the system's Lagrangian, $L = T - V$, where T is the total kinetic energy and V is the total potential energy.

2.2.1 Nonlinear Equations of Motion

By applying the Euler-Lagrange equation to the generalized coordinates θ and α , the nonlinear equations of motion for the system are obtained:

$$\begin{aligned} (m_p L_r^2 + m_m L_{arm}^2 + \frac{1}{2} m_p L_p^2 + \frac{1}{2} m_m L_p^2 - \frac{1}{2} m_p L_p^2 \cos(2\alpha) - \frac{1}{2} m_m L_p^2 \cos(2\alpha) + J_{arm}) \ddot{\theta} \\ - (m_p L_p L_{arm} \cos(\alpha) + m_m L_p L_{arm} \cos(\alpha)) \ddot{\alpha} + (m_p L_p^2 \sin(2\alpha) + m_m L_p^2 \sin(2\alpha)) \dot{\theta} \dot{\alpha} \\ + (m_p L_p L_{arm} \sin(\alpha) + m_m L_p L_{arm} \sin(\alpha)) \dot{\alpha}^2 = \tau - B_{arm} \dot{\theta} \end{aligned} \quad (1)$$

$$\begin{aligned} - (\frac{1}{2} m_p L_p L_r \cos(\alpha) + \frac{1}{2} m_m L_p L_r \cos(\alpha)) \ddot{\theta} + (J_p + m_p L_p^2 + m_m L_p^2) \ddot{\alpha} \\ - (\frac{1}{2} m_p L_p^2 \sin(2\alpha) + \frac{1}{2} m_m L_p^2 \sin(2\alpha)) \dot{\theta}^2 - m_p L_p g \sin(\alpha) - m_m L_p g \sin(\alpha) = -B_p \dot{\alpha} \end{aligned} \quad (2)$$

Where τ is the torque applied by the DC motor at the base of the rotary arm, defined as:

$$\tau = \frac{\eta_g K_g \eta_m k_t (V_m - K_g k_m \dot{\theta})}{R_m} \quad (3)$$

2.2.2 Linearization

To design a linear state-feedback controller, the system must be linearized around the upright equilibrium point ($\alpha = 0$). Assuming small perturbations, the following approximations are used:

- $\sin(\alpha) \approx \alpha$
- $\cos(\alpha) \approx 1$

- $\dot{\alpha}^2 \approx 0, \dot{\theta}^2 \approx 0, \dot{\theta}\dot{\alpha} \approx 0$ (Higher-order terms are neglected)

Applying these simplifications results in the linearized equations of motion in terms of the mass/inertia matrix M , the damping matrix B , and the stiffness/gravity matrix K :

$$M\ddot{q} + B\dot{q} + Kq = \text{RHS} \cdot V_m \quad (4)$$

The vector $\text{RHS} = [K_v, 0]^T$ represents the input coupling of the system, where K_v is the actuator gain that translates the control voltage V_m into a generalized force applied to the rotary arm, while the zero entry reflects that the pendulum link is un-actuated.

2.3. State-Space Representation

To isolate the acceleration terms ($\ddot{q}$), the system is multiplied by the inverse of the mass matrix (M^{-1}):

$$\ddot{q} = -M^{-1}B\dot{q} - M^{-1}Kq + M^{-1}\text{RHS} \cdot V_m \quad (5)$$

In the implementation code, these intermediate terms are calculated as:

- $A_1 = -M^{-1}B$
- $A_2 = -M^{-1}K$
- $B_1 = M^{-1}\text{RHS}$

The linearized equations of motion are rearranged into the standard first-order state-space form:

$$\dot{x} = Ax + Bu \quad (6)$$

$$y = Cx + Du \quad (7)$$

2.3.1 Definition of States, Inputs, and Outputs

For the rotary inverted pendulum system, the state vector x is defined to capture the position and velocity of both degrees of freedom:

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \\ \alpha \\ \dot{\alpha} \end{bmatrix} \quad (8)$$

where θ and α are the arm and pendulum angles, and $\dot{\theta}$ and $\dot{\alpha}$ are their respective angular velocities. The control input u is the motor voltage V_m .

The output vector y tracks the measured angles:

$$y = \begin{bmatrix} \theta \\ \alpha \end{bmatrix} \quad (9)$$

2.3.2 System Matrices

The C and D matrices are defined based on the sensors available:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (10)$$

The A and B matrices contain the dynamics derived from the mass (M), damping (B_{damp}), and stiffness (K) matrices. By solving for the accelerations, the rows of A and B are populated as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ A_{2(1,1)} & A_{1(1,1)} & A_{2(1,2)} & A_{1(1,2)} \\ 0 & 0 & 0 & 1 \\ A_{2(2,1)} & A_{1(2,1)} & A_{2(2,2)} & A_{1(2,2)} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ B_{1(1)} \\ 0 \\ B_{1(2)} \end{bmatrix} \quad (11)$$

Note that rows 1 and 3 of the A matrix simply define the relationship $\dot{x}_1 = x_2$ and $\dot{x}_3 = x_4$, while rows 2 and 4 contain the physics of the system acceleration.

2.4. Control Concepts Used

The successful operation of the rotary inverted pendulum requires a combination of feedforward optimal control with LQR for the Swing-Up part and state feedback law for linear stabilization with a Kalman filter to estimate unmeasured states.

2.4.1 Balance Control: Pole Placement

Pole placement is a state-feedback technique used to stabilize the system once the pendulum is in the near-upright position.

The natural dynamics of the pendulum are defined by the eigenvalues of the system matrix A . In its open-loop state, the inverted pendulum is unstable because at least one eigenvalue exists in the Right-Half Plane (RHP).

A control law $u = -Kx$ is defined. By choosing the gain matrix K , the closed-loop system $\dot{x} = (A - BK)x$ is forced to have eigenvalues (poles) at specific locations in the Left-Half Plane (LHP).

The poles are "placed" to satisfy specific requirements for damping ($\zeta = 0.85$) and natural frequency ($\omega_n = 4$ rad/s), ensuring the arm corrects the pendulum's position quickly without excessive oscillation.

2.4.2 Swing-Up Control: Optimal Control (Indirect Method)

The swing-up phase is formulated as a trajectory optimization problem that transfers the pendulum from the stable hanging equilibrium to the unstable upright position. An indirect optimal control approach is employed, where the Pontryagin Maximum Principle is applied to derive the coupled state and co-state dynamics.

The resulting two-point boundary value problem is solved numerically using a steepest descent method, with the state and co-state equations integrated using the MATLAB `ode45` solver. The co-state dynamics are integrated backward in time from the desired final state, providing the gradient required to iteratively update the control input until convergence to an optimal swing-up trajectory.

Once the optimal path is calculated, it serves as a reference. A Linear Quadratic Regulator (LQR) is utilized to calculate the time-varying feedback gains required to correct the system's actual position relative to this optimal path. LQR ensures the system stays "on track" by minimizing a quadratic cost function that weights state deviations against the control effort.

2.4.3 State Estimation: Kalman Filtering

To implement the pole placement gains, the full state vector $x = [\theta, \dot{\theta}, \alpha, \dot{\alpha}]^T$ must be known at every time step. Since the encoders only directly provide θ and α measurements, a Kalman Filter is used to estimate the angular velocities.

Without a Kalman Filter, velocity would be calculated by simply taking the derivative of the position signal ($\Delta\theta/\Delta t$). Because encoders have finite resolution, this direct differentiation introduces significant high-frequency noise. This noise can lead to motor behavior and system instability.

The Kalman Filter uses the system's A and B matrices to predict the next state, then corrects that prediction with the actual sensor data.

By weighting the physics-based model against the noisy measurements, the KF provides a "clean" estimate of $\dot{\theta}$ and $\dot{\alpha}$. This allows for smooth control actions and prevents the actuator from reacting to sensor noise, which is critical for the high-speed transitions required during the swing-up and balance phases.

3. System Identification and Parameter Extraction

The primary objective of this section is to determine an accurate mathematical model for the entire robotic system. Given the complexity of the multi-body dynamics, the identification process begins with the parameter estimation of a simplified 1 DOF system. By isolating the primary joint, it is possible to precisely characterize the damping and inertia of the arm that govern the system's behavior before extending the model to the full system dynamics.

3.1. Theoretical Parameter Calculation

Before performing experimental identification, a baseline model was established using datasheet specifications.

3.1.1 Mechanical Domain

The total moment of inertia (J) and the viscous damping coefficient (B) are the primary mechanical parameters. These are calculated by reflecting the properties of the load (the arm) and the actuator to the output shaft.

- **Total Moment of Inertia (J):** Calculated as the sum of the arm's inertia and the equivalent motor inertia:

$$J = J_{arm} + J_{eq} \quad (12)$$

where J_{arm} is derived from the mass and geometry of the link (datasheet value), and J_{eq} represents the rotor inertia scaled by the gear ratio.

- **Passive Damping ($B_{disabled}$):** This accounts for the physical friction in the bearings when the motor is not powered:

$$B_{disabled} = B_{arm} + B_{eq} \quad (13)$$

3.1.2 Electromechanical Domain

The coupling between the electrical input (voltage V) and the mechanical output (velocity $\dot{\theta}$) is defined by the motor's torque constant (k_t), back-EMF constant (k_m), and the armature resistance (R). Taking into account the gear ratio (K_g) and the efficiencies (η_m, η_g), the effective parameters are:

- **Electromechanical Damping (B):** The total damping includes the "back-EMF damping" effect, which provides a resistive torque proportional to velocity:

$$B = B_{arm} + B_{eq} + \frac{\eta_m \eta_g K_g^2 k_t k_m}{R} \quad (14)$$

- **Voltage Gain (K_v):** This parameter defines the steady-state torque production per unit of input voltage:

$$K_v = \frac{\eta_m \eta_g K_g k_t}{R} \quad (15)$$

3.1.3 State-Space Representation

Using these derived parameters, the 1 DOF velocity model is expressed in state-space form as:

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{B}{J} \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{K_v}{J} \end{bmatrix} V \quad (16)$$

3.2. Data Extraction and Preprocessing

The first step in the identification process is to convert raw sensor data into a structured dataset suitable for analysis. This was achieved using a custom `extract_data` script, which allows for windowing and extraction of multiple tests while also filtering noisy estimates from a single Simulink log.

Once the test segments were cleaned and isolated, a numerical optimization was performed to refine the system parameters.

3.2.1 Optimization Strategy

The identification process is framed as a "matching" problem. The optimal values for the motor gain (K_v) and the damping coefficient (B) are sought that allow the mathematical model to best replicate the physical behavior observed in the lab.

The optimization starts with initial estimates (K_{v0}, B_0, J_0) derived from theoretical datasheets. This provides a physically meaningful starting point for the algorithm.

A function `global_model_err` was defined to calculate the cumulative error across all test segments. For each iteration, the script uses the `lsim` command to simulate the model's response to the measured experimental voltage.

The `fminsearch` algorithm was used to minimize the Sum of Squared Errors (SSE) between the simulated velocity (ω_{sim}) and the experimental velocity (ω_{exp}):

$$J(\theta) = \sum_{k=1}^{num_tests} (\dot{\theta}_{exp,k} - \dot{\theta}_{sim,k})^2 \quad (17)$$

3.2.2 Final Fit Results

During this process, the moment of inertia J was held constant, and the final "Fitted" values for K_v and B were extracted once the algorithm converged.

As shown in the resulting plots, the fitted model (red dashed line) accurately tracks the experimental data (blue line) for both positive and negative steps. Close alignment in the rise time and the final velocity level confirms that the identified parameters are accurate for use in the control design phase.

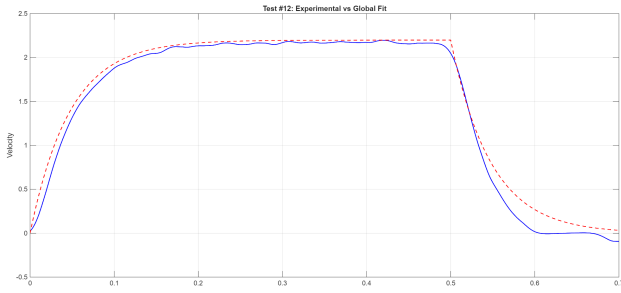


Figure 3: Experimental vs Analytical Model Fit just with arm (Test 12-positive step)

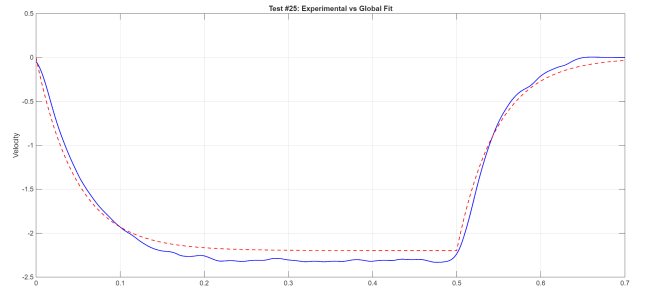


Figure 4: Experimental vs Analytical Model Fit just with arm (Test 25-negative step)

3.3. 2-DOF Pendulum Parameter Estimation

With the arm parameters already identified, the final step in the system identification is to characterize the pendulum dynamics.

The pendulum's local moment of inertia J_p was determined using the theoretical model for a slender rod. While the pendulum rotates about its pivot (the tip), the dynamic equations are derived relative to the Center of Mass (CoM). Therefore, J_p is defined as:

$$J_p = \frac{1}{12} m_p L_{rod}^2 \quad (18)$$

The total rotational inertia about the pivot is correctly accounted for in the system's mass matrix ($M_{22} = J_p + m_p l_p^2$) via the Parallel Axis Theorem, where l_p is the distance to the center of mass.

3.3.1 Fixed vs. Estimated Parameters

To maintain physical consistency and avoid over-parameterization, several constants were held fixed based on physical measurements and the theoretical 1-DOF results:

J_{tot} , B_{tot} , and K_v were kept at the values identified in the previous 1-DOF identification stage. The lengths L_{arm} , l_p , l_m and rodlength and the masses $m_{pendulum}$ and m_{metal} were measured directly from the physical system.

3.3.2 Nonlinear Optimization for B_p

The parameter B_p was estimated by solving a nonlinear least-squares problem. The solver, `lsqnonlin`, was configured to minimize the difference between the experimental and simulated state vectors:

$$\min_{B_p} \sum_{k \in tests} \|\mathbf{x}_{exp,k} - \mathbf{x}_{sim,k}(B_p)\|_W^2 \quad (19)$$

Where W is a weighting vector that prioritizes the residuals of the pendulum angle (α) and velocity ($\dot{\alpha}$). This "Global Fit" approach stacks data from multiple tests into a single cost function, ensuring the resulting B_p is robust across different initial conditions and input voltages.

3.3.3 Validation and Fit Results

The following figures show the comparison between the identified non linear model and the experimental one.

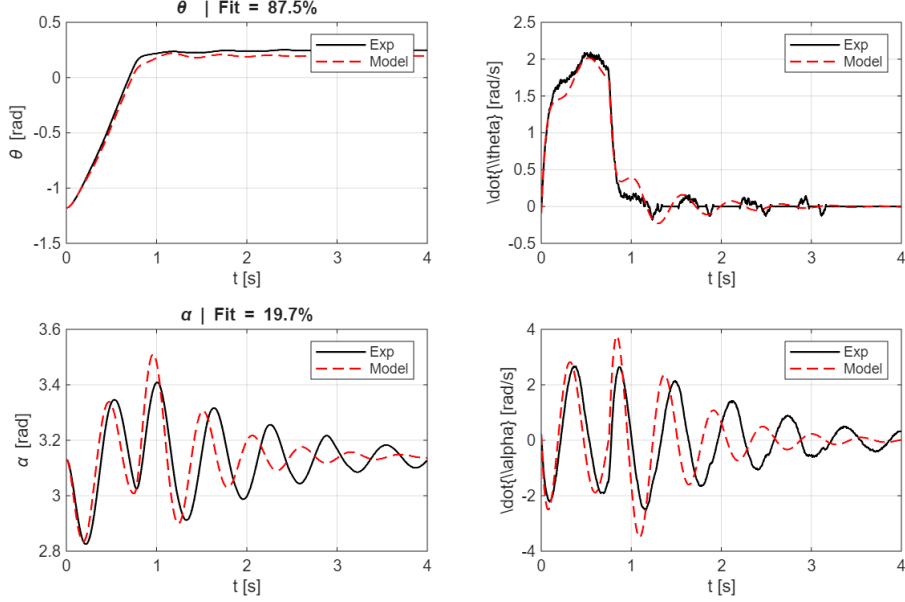


Figure 5: Experimental vs Analytical Model Fit with arm and pendulum (Test 17 - positive step)

As the pendulum stops being actuated, the mechanical tension from the electrical wiring becomes a dominant unmodeled disturbance, preventing the simulation from perfectly matching the experimental amplitude decay.

However, because the primary objective of the model is to provide a basis for the LQR and Pole Placement controllers near the upright equilibrium—where velocities and control forces are active—the parameters derived from the initial actuated phase are sufficient to ensure system stability and performance in the final implementation.

Therefore the estimated value for damping coefficient B_p remains valid for control design.

3.4. Summary of Identified Parameters

The finalized parameters used for the control design and linearization are summarized in the table below.

Parameter	Symbol	Value	Unit
Total Arm Inertia	J_{tot}	4.08e-3	kg·m ²
Total Arm Damping	B_{tot}	8.58e-2	N·m·s/rad
Motor Gain	K_v	7.86e-2	N·m/V
Pendulum Inertia (CoM)	J_p	6.68e-5	kg·m ²
Pendulum Damping	B_p	1.0036e-4	N·m·s/rad

Table 1: Identified physical constants for the 2-DOF robotic system.

4. Computation and Simulation

In order to optimize the workflow and reduce reliance on limited laboratory access and enable the testing of different concepts, a digital twin of the pendulum was implemented in MATLAB. Furthermore, this simulation served as the primary test subject for validating and tuning the control efforts, as well as corroborating the fidelity of the identified model.

4.1. Model Implementation

4.1.1 Non-Linear Dynamics

The core of the simulation model consists of a nonlinear plant implementing the full Euler–Lagrange equations derived in Section 2.2. Due to the strongly nonlinear nature of the system, a fixed linear state-space representation is insufficient, requiring the full multi-DOF dynamics to be evaluated at each simulation step. This allows the accurate capture of effects such as Coriolis and centripetal forces, which are particularly relevant during the fast swing-up motion.

At each simulation step, the MATLAB Function block reconstructs the configuration-dependent mass matrix $M(q)$ and the vector of nonlinear terms $C(q, \dot{q})$ based on the current state vector

$$x = [\theta, \dot{\theta}, \alpha, \dot{\alpha}]^T.$$

The joint accelerations are obtained by solving

$$\ddot{q} = M(q)^{-1} (\tau_{\text{net}} - C(q, \dot{q})),$$

where τ_{net} includes the motor torque and the identified viscous friction terms. This formulation avoids linearization errors associated with small-angle approximations and ensures model validity over the full operating range of the pendulum.

A 3D animation was also implemented to qualitatively assess the fidelity of the digital twin.

4.1.2 Realistic modeling of hardware constraints and disturbances

To increase the fidelity of the digital twin with respect to the real system, several nonlinearities, system constraints, and pseudo-random processes were introduced into the simulation. This allowed for preliminary controller tuning, robustness validation, and a reduction in adaptation effort when transferring the methodologies to the real system, enabling a more efficient laboratory workflow. The extent of this realistic modeling included:

- **Actuator Saturation:** A saturation block was placed before the plant input to model the physical voltage limits of the DC motor (12 V). This ensures that the implemented control strategies, particularly the swing-up controller, do not request infeasible voltages.
- **Process Noise:** To assess the robustness of the feedback controller and LQR against unmodeled dynamics, band-limited white noise was injected into the plant inputs, representing stochastic disturbances such as unmodeled friction and residual cable tension.
- **Sensor Noise:** Gaussian white noise was added to the plant output measurements to simulate encoder resolution limits and electrical noise observed in the real system.
- **Discrete-Time Sampling:** Zero-order hold (ZOH) blocks were used in the feedback loop to enforce the discrete sampling interval of the real-time controller and capture discretization and delay effects.

4.2. Swing-Up Trajectory Optimization

Rather than implementing an energy-based swing-up procedure, which injects energy into the system based on the pendulum’s natural swinging behavior, the swing-up maneuver was formulated as a constrained optimization problem. This choice was primarily driven by two physical constraints of the real system:

1. **State Constraints:** The most relevant mechanical limitation is the restricted range of motion of the rotary arm due to sensor cabling. Excessive arm rotation would introduce unmodeled drag from cable stretching and risk damaging the connections. Consequently, a constraint on the arm angle θ was imposed, limiting its motion to a $[-35^\circ, 35^\circ]$ range relative to the neutral position.
2. **Actuator Saturation:** Energy-based methods do not explicitly account for actuator limitations and may request voltages exceeding the DC motor capabilities, particularly during fast transients. By adopting an optimal control framework, the control effort can be penalized to ensure the requested torque remains within the actuator saturation limits.

Since compliance with both constraints cannot be guaranteed using a simple energy-based approach, the optimal reference trajectory was computed offline using the indirect method described in Section 2.4.2.

4.2.1 Cost Function Implementation

In order to translate the aforementioned theoretical framework and physical constraints into a solvable numerical problem, the cost function J was tuned to prioritize the end goal over path efficiency. The optimization minimizes:

$$J = \frac{1}{2}(x(t_f) - x_f)^T P_f (x(t_f) - x_f) + \int_a^b \left(\frac{1}{2} u^T R u + \frac{1}{2} x^T Q x \right) dt \quad (20)$$

The weighting matrices were normalized to their maximum expected values, namely voltage ($V_{\max} = 12$ V), angles ($\theta_{\max} = 35^\circ$, $\alpha_{\max} = 360^\circ$), and a similar approach to velocities, effectively serving as soft constraints for the optimization procedure.

4.2.2 Numerical Solver Configuration

The numerical solution was computed using the gradient descent algorithm described in Section 2.4.2. To ensure stability during the iterative process, the `ode45` solver tolerances were tightened to 10^{-4} . Additionally, the descent step size was manually tuned to a conservative value ($\eta = 0.0005$) to prevent the optimizer from overshooting the cost function minima, a behavior that was observed to cause divergence with larger step sizes.

4.3. Finite-Horizon LQR Tracking

To guarantee robust performance of the swing-up against model mismatches, a Linear Quadratic Regulator (LQR) was implemented via the "Gain Scheduling" block, with the feedback gains $K(t)$ being computed offline by backwards integration of the Differential Riccati Equation.

4.3.1 Terminal State Emphasis

A high terminal state weight was defined, which translated into higher magnitude gains in the later parts of the maneuver, enforcing the importance of accurately reaching the final state ($P(t_f) = P_f$) over strict path following throughout the procedure.

This desired behavior can be seen in the computed LQR gain magnitude graph bellow:

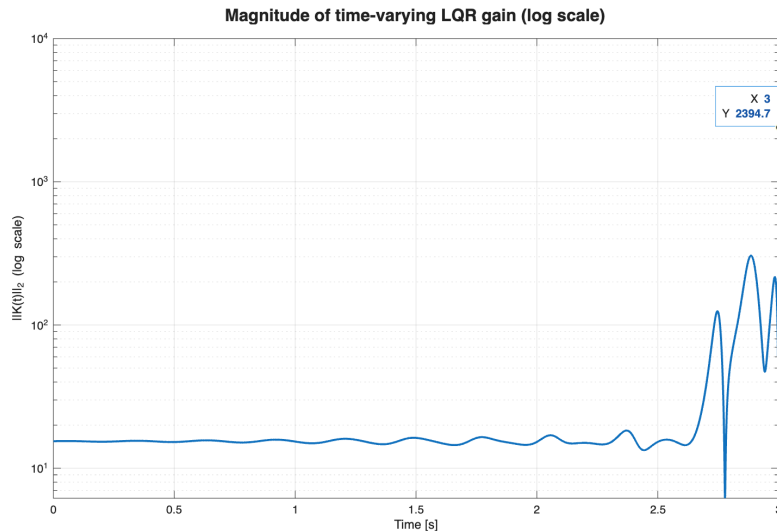


Figure 6: Magnitude of time-varying LQR gain (log scale)

4.3.2 LQR non-filtered Feedback

Due to this process occurring in a highly non-linear environment, the state feedback given to the LQR wasn't provided from the Linear Kalman Filter, but rather with direct position and their filtered derivatives from the plant.

4.4. Hybrid Control Logic

A simple switch logic was defined to manage the transition between the non-linear swing-up routine and the linearized stabilization phase, which is triggered when the angular position of the pendulum is detected to be within the range for the linearized system to be admissible and thus allowing the feedback control to take over.

$$u(t) = \begin{cases} u_{\text{swing-up}}(t) & \text{if } |\alpha| > 20^\circ \\ -Kx(t) & \text{if } |\alpha| \leq 20^\circ \end{cases}$$

4.5. Feedback Control at the unstable Equilibrium Point

4.5.1 Pole placement using state feedback

Given the defined reduced operating window $\alpha \in [-20^\circ, 20^\circ]$ of the control architecture, without losing too much precision, it is possible to linearize the non-linear dynamics around the $\alpha = 0^\circ$ equilibrium position. This simplifies the complexity of the problem and allows for the employment of linear control techniques.

As introduced in Section 2.4.1, the control of the pendulum in the vicinity of the unstable upward position was carried out through state-feedback control and pole location defined via the pole placement method, which allows for the tuning of the behavioral characteristics of the system, namely $w_n = 4$ rad/s and $\zeta = 0.85$.

The estimated state vector computed by the Kalman Filter (refer to Section 4.6) is thus multiplied by the offline calculated static gain vector and fed back into the plant (given that a null reference state vector is considered).

4.5.2 Stabilization with Integral Action

Standard state-feedback regulation often results in steady-state errors due to unmodeled friction or constant disturbances. To mitigate this, the stabilization controller was augmented with an integral action on the arm position error, which also allows for the change in the reference arm final state and thus enabling the movement to whichever position is desired.

The definition of this error is as follows: $e_\theta = \theta_{\text{ref}} - \theta$, which translates to the formulation of the additional state as $x_I = \int (\theta_{\text{ref}} - \theta) dt$. This results in the feedback law being expanded to:

$$u_{\text{ext}} = -K_x x(t) - K_I \int_0^t (\theta_{\text{ref}} - \theta(\tau)) d\tau$$

where the integral gain K_I corresponds to the last entry of the extended gain vector K_e .

4.6. State Estimation via Kalman Filtering

To implement the state feedback control methodology, as previously mentioned, the full state vector is required, featuring both the angular position of the arm and pendulum, but also their respective velocities. While these could be computed via numerical differentiation, as is done by the embedded controller of the encoders, this approach is highly susceptible to noise amplification as mentioned in Section 2.4.3.

To mitigate this, a Discrete-Time Kalman Filter was implemented to reconstruct the full state vector by considering both the encoders' measurements and the system model.

4.6.1 Discrete-Time Implementation

The estimator is implemented into the control loop and works at the sampling time at which measurements are provided from the plant. Thus, the discrete time linearized matrices (A_d, B_d) are employed to predict the state evolution dynamics between sampling instances and create a trustful full-state vector that can then be used by the control architecture.

4.6.2 Covariance Tuning Strategy

The tuning of the key parameters of the estimator relies on the relative balancing of the expected process noise and measurement noise covariances (Q, R), which can be interpreted as the trust given to each of those occurrences.

Rather than solely relying on arbitrary tuning, the values attributed to these matrices were defined to reflect the expected behavior of the hardware, as well as to account for unmodeled incidences, with the latter finetuning being done iteratively.

- **Measurement Confidence (R):** The measurement covariance matrix was initially determined according to the expected variance of the encoders solely due to limited resolution, thus creating an adequate baseline for the tuning efforts.

Given an encoder resolution of $N = 4000$ counts per revolution, the angular resolution is defined as $\Delta = 2\pi/N$. Modeling the error as a uniform distribution with Δ width, the variance σ^2 is calculated as:

$$R = \text{diag}(\sigma_\theta^2, \sigma_\alpha^2), \text{ where } \sigma^2 = \frac{\Delta^2}{12} \approx 2.06 \times 10^{-7} \text{ rad}^2 \quad (21)$$

This represents the minimum admissible variance. To account for the real stochastic processes affecting the real system, this matrix was iteratively fine tuned on site by multiplication of a scalar α , which turned out to be ≈ 5 .

- **Process Disturbance (Q):** The process noise covariance was modeled to account for unmodeled dynamics presenting themselves as friction or torque anomalies, thus this matrix was structured to put less reliance on the velocity terms $(\dot{\theta}, \dot{\alpha})$ and allowing for adaptation to these external disturbances while trusting the remaining model.

4.6.3 Estimator Validation

The simulation developed enabled the validation of the Kalman Filter by comparing the estimated state vector against the model's ideal states, allowing for the validation of the filter's convergence and stability before any actual physical implementation.

4.6.4 Extended Kalman Filter Implementation

In order to augment the operating window of the estimator past the linearized equilibrium zone, an extended version of the Kalman Filter was designed, allowing for full-state estimation throughout the complete control process. However, while this model was successfully implemented in the digital twin, there were complications relating to the deployment of this tool in the real system, which ultimately prevented its practical use. This will be further discussed in Section 6.

5. Experimental Results and Discussion

This section presents the experimental validation of the proposed control strategy on the rotary inverted pendulum system.

5.1. Experimental Setup and Test Conditions

The experimental setup consists of a horizontally actuated arm driven by a DC motor and a pendulum freely rotating in the vertical plane. Angular positions of the arm and pendulum are measured using rotary encoders. All experiments were conducted using a fixed sampling period of $T_s = 0.004$ s. Actuator voltage and current saturation limits were enforced to ensure safe operation of the system. The control strategies described in Section 4 were implemented without modification on the real system, including the swing-up controller, the stabilization controller around the upright equilibrium, and the linearized Kalman filter for state estimation, with the only difference being the process and measurement noise covariance values.

5.2. Experimental Test Scenarios

Several experimental scenarios were conducted to evaluate the performance of the control system under real-world conditions. These scenarios were designed to assess swing-up capability, stabilization performance, and robustness to disturbances.

The considered test cases include:

- Swing-up from the downward equilibrium
- Capture and stabilization around the upright unstable equilibrium
- Response to external disturbances applied during stabilization

- Achieving zero error regulation with reference arm angle

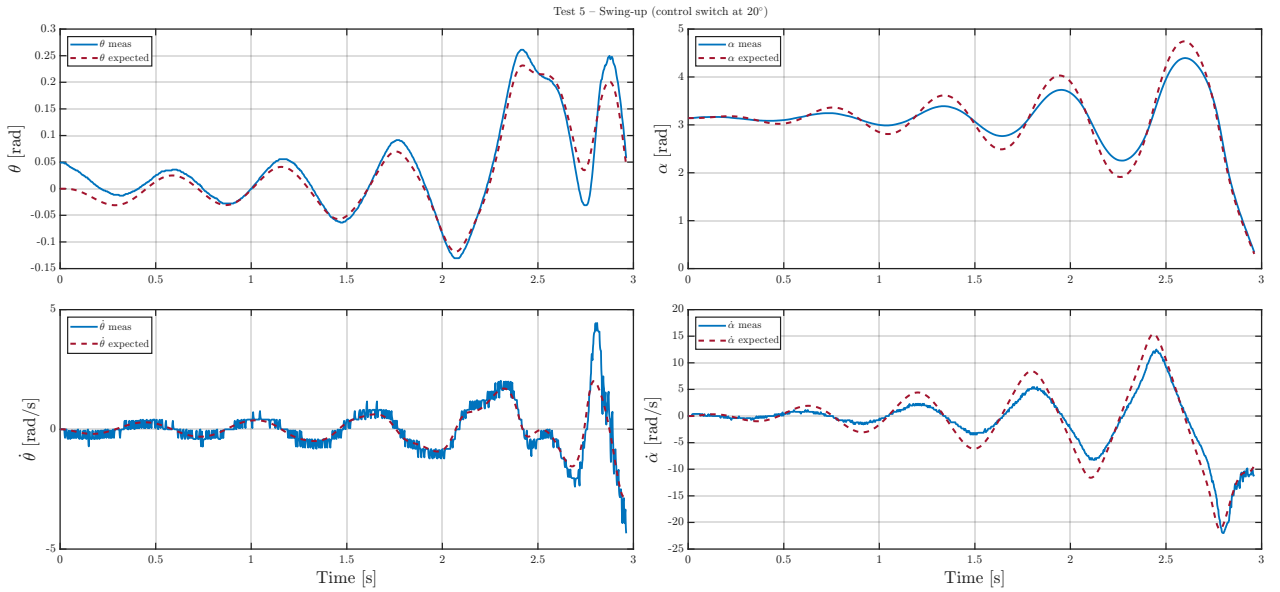
5.2.1 Experimental Results: Swing-Up with Feedforward Optimal Control w/ LQR

This subsection presents the experimental results obtained during the swing-up phase. Starting from the downward equilibrium, the feedforward optimal control with neighboring LQR controller successfully manages to drive the system to its upward position.

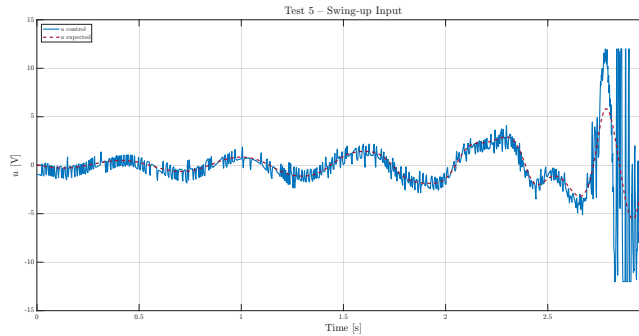
The experimental results demonstrate consistent swing-up behavior across multiple trials. An example of the state evolution together with the expected state evolution and the corresponding control inputs is shown in Fig. 7.

It can be observed that the overall state evolution matches the expected trajectory well. In particular, the initial difference in the arm angle θ is gradually corrected toward the expected evolution due to the LQR control. Although the states deviate from the expected trajectory during the motion, they match almost exactly near the end of the swing-up.

This behavior is expected for a finite-horizon LQR controller. As discussed in the previous sections on LQR, the feedback gains increase as the system approaches the final time due to the high terminal cost definition. As a result, state deviations are corrected with higher control effort near the completion of the swing-up, causing the states to align closely with the expected behavior.



(a) Swing-up states (measured vs expected).



(b) Swing-up input (measured u vs u^*).

Figure 7: Swing-up phase results for the rotary inverted pendulum.

5.2.2 Experimental Results: Stabilization Around the Upright Equilibrium

Once the pendulum enters the vicinity of the upright equilibrium, the stabilization controller is activated. The experimental response for a single test during this phase is shown in Fig. 8.

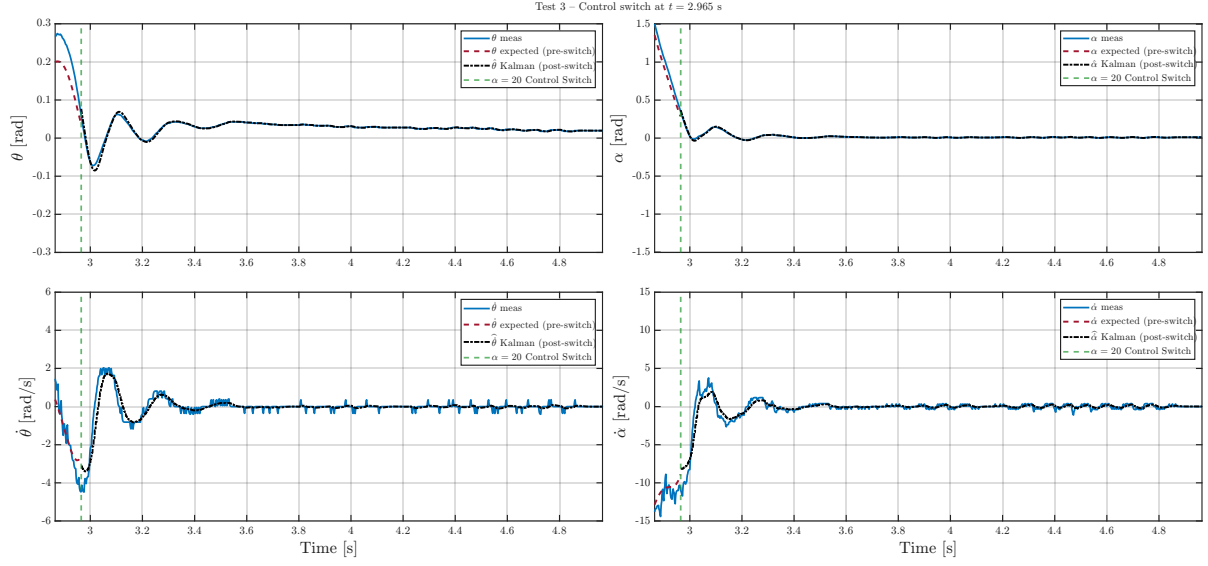
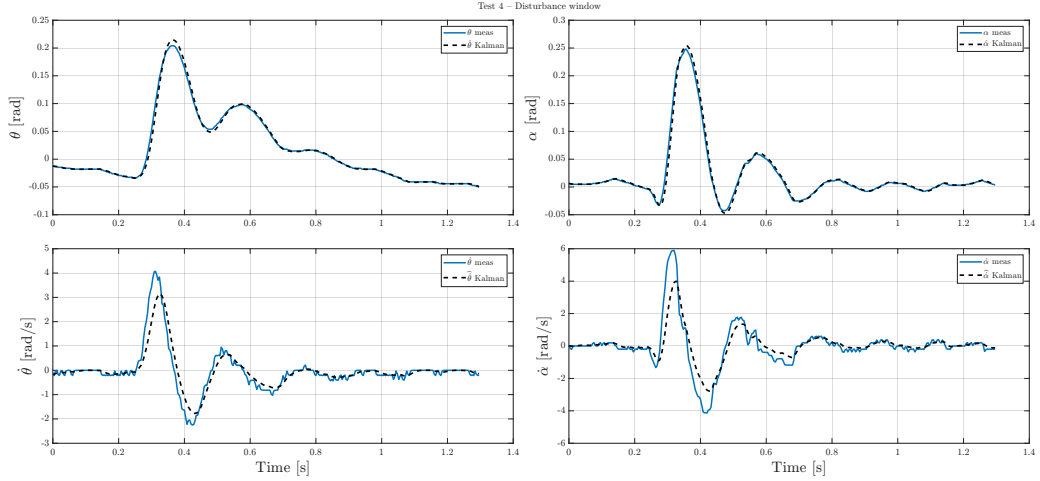
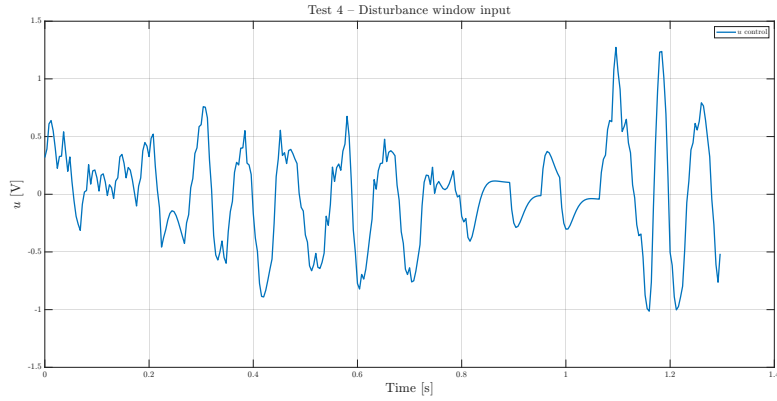


Figure 8: Stabilization Around Unstable Equilibrium After Control Switch

After switching to state feedback, the system achieves stable regulation around the unstable equilibrium with decaying oscillations. It is important to note that the decaying oscillations are not representative of the expected dominant pole used for the pole placement. This issue will be discussed in later sections. The system also shows proper response to external disturbances, and achieves stability after being disturbed. An example system response is provided in 9.



(a) States during external disturbance.



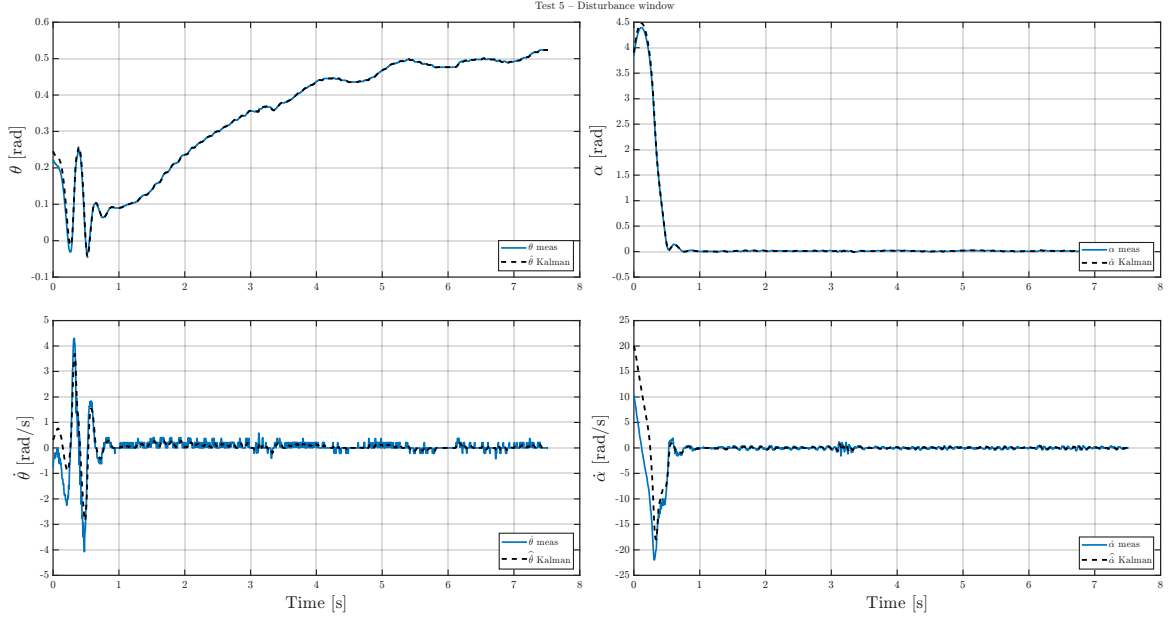
(b) Control input.

Figure 9: Disturbance response of the system.

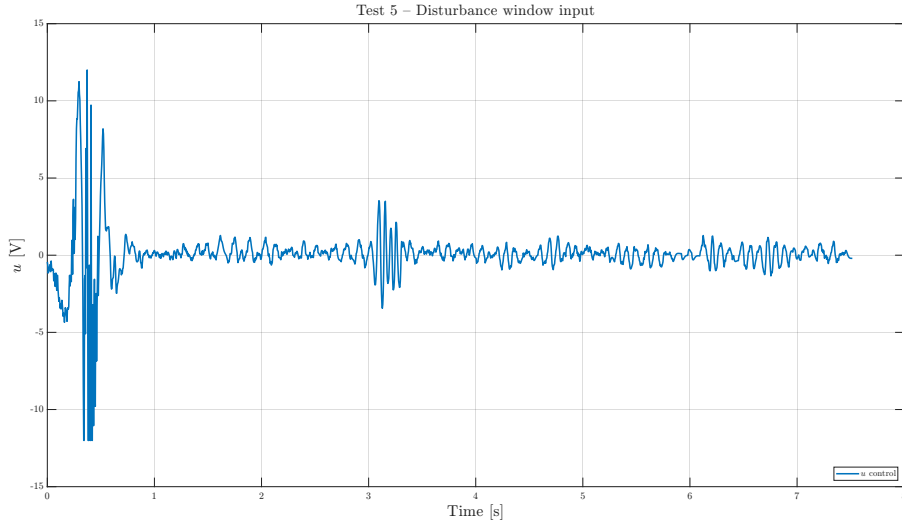
5.2.3 Experimental Results: Response to a Arm Angle Reference

The system was also tested with a reference input to validate the extended system pole placement. The pole of integral error was placed at $\sigma = -0.5$, which is way slower than the previous dominant poles of $p = -3.4000 \pm 2.1071i$. The resulting eigenvector for the newly placed pole is $[-0.4364; 0.2182; 0.0024; -0.0012; -0.8729]$. Considering the state vector 8, it can be expected that the response regarding the error will largely affect the states related with the arm whereas pendulum dynamics won't be affected by it.

The state evolutions and control input for the case of reference arm angle of $\theta_r = 30deg = 0.52rad$ are provided in 10. The system behaves as expected where the pendulum holds stability, while the state θ approached the reference with a transient decay behaviour which is in line with the pole placed, where the expected settling time $t_s = 4/\sigma = 8s$ is also apparent in the state θ



(a) States during external disturbance.



(b) Control input.

Figure 10: Stabilization with arm reference angle

5.2.4 State Estimation Performance

The performance of the linearized Kalman filter was evaluated by comparing the estimated states against the measured signals. As shown in Figure 8, the filter effectively attenuates measurement noise, providing smooth state estimates suitable for feedback control.

To quantify the estimation performance, the averaged cross-correlations for each state across four tests are presented in Figure 11. The analysis confirms that the position estimates are highly correlated with the measurements. The velocity estimates exhibit slightly lower correlation, which is expected given the inherently noisy nature of velocity measurements.

Regarding estimation latency, the position estimates lag the measurements by approximately one sample, while the arm and pendulum velocity estimates show a lag of approximately three and four samples, respectively. While an ideal Kalman filter should propagate states in perfect synchronization with the actual system evolution, this implementation does not consider velocity measurements instead it, derives them through positions measurements which explains the higher delay in velocity measurements, additionally delays are often introduced by model mismatches or the tuning of noise covariance matrices. Despite these minor delays, the estimations remain well within the acceptable range for this application.

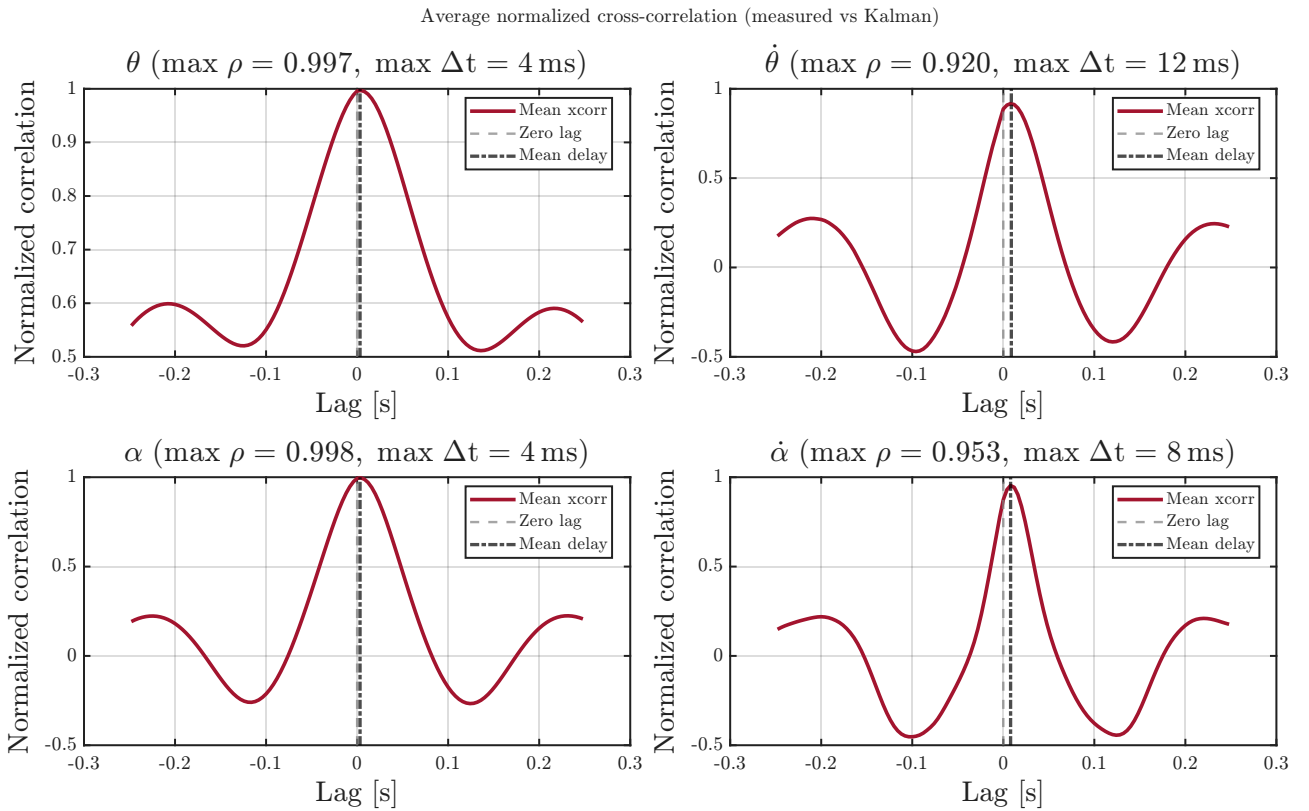


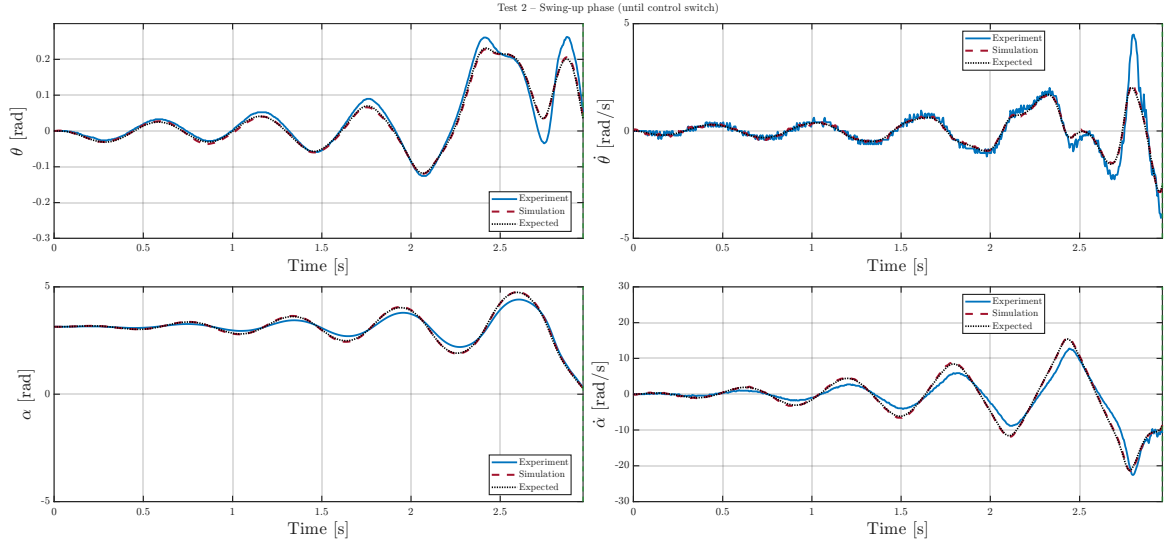
Figure 11: Estimates and Measurements Averaged Cross Correlation Results

5.3. Simulation and Experimental Comparison

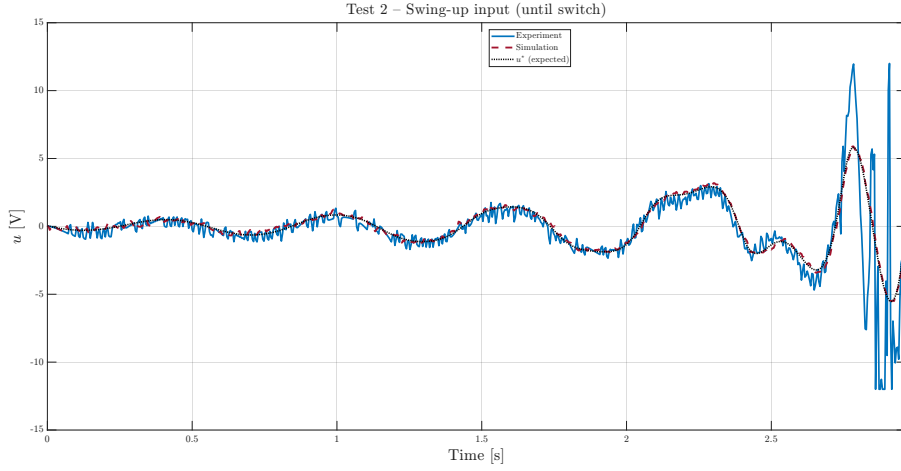
Although the controller was designed and validated in simulation, this section focuses on comparing experimental behavior with simulation results.

5.3.1 Swing-Up Behavior

An example swing-up test was compared with the simulation outputs in 12. As expected, the swing-up portion of the simulation matches the optimal state evolution exactly, despite the presence of random torque disturbances modeled as acceleration noise. This is because the simulation represents an idealized case in which the optimal control model and the simulated system are perfectly matched. In contrast, the experimental results follow the predicted state evolution closely, with deviations remaining within an acceptable range. This agreement confirms that the system modeling, optimal control trajectory, and LQR controller were implemented correctly.



(a) State evolution comparisons of experiment and simulation.



(b) Control inputs of experiment and simulation..

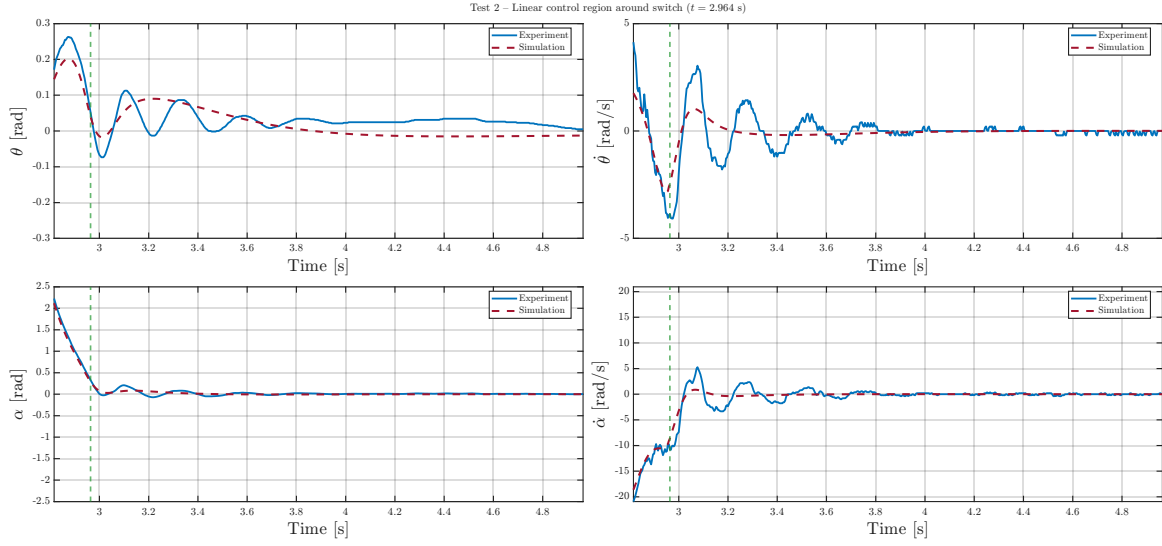
Figure 12: Comparison of the swing-up phase.

5.3.2 Stabilization Performance

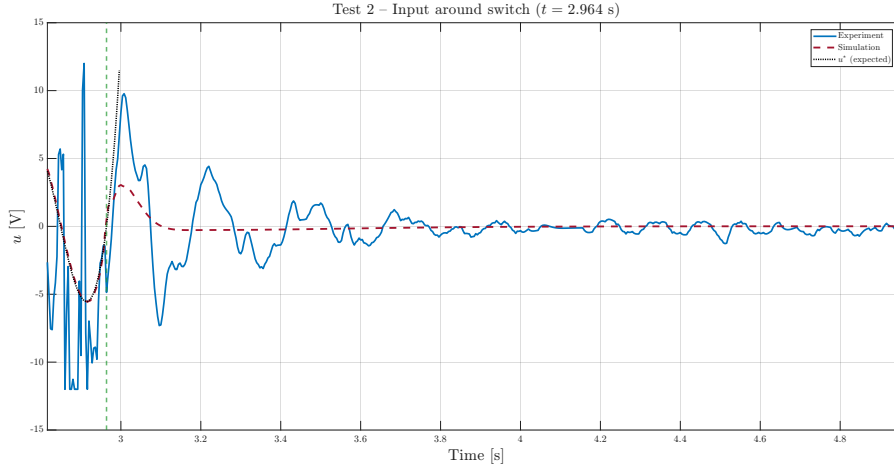
In Figure 13, the linear-region stabilization behavior is compared between the simulation (without disturbance noise) and the experimental results. The simulation exhibits the expected behavior associated with the placed closed-loop poles, resulting in an almost critically damped transient response during stabilization. In contrast, the experimental response reveals a different dominant pole than predicted by the design.

It is important to note that, in this operating region, nonlinear effects such as actuator dead-zone and backlash become more pronounced due to the lower-frequency switching of the control voltage, which leads to intermittent arm rotations.

Nevertheless, the controller maintains stability within the valid linearization region and successfully achieves stabilization.



(a) State evolution comparisons of experiment and simulation.



(b) Control inputs of experiment and simulation..

Figure 13: Stabilization with arm reference angle

5.4. Summary of Experimental Results

The results indicate that the proposed control architecture is capable of stabilizing the pendulum under real-world conditions, despite the presence of nonlinearities and uncertainties not captured in the model. Observed discrepancies between experimental and simulated responses highlight the influence of unmodeled dynamics such as backlash, friction, and actuator saturation. These effects motivate the consideration of more detailed modeling or adaptive control strategies in future work.

6. Shortcomings

As previously mentioned throughout the report, there were some major shortcomings that will be here further addressed:

Stabilization exhibited decaying oscillations inconsistent with the designed dominant poles, likely due to unmodeled actuator dead-zones and backlash.

Deployment of the EKF was prevented by Simulink-to-C compilation errors that are suspected to be a bit resolution issue, restricting the filter to the linear region and forcing noisy feedback during swing-up.

The low accuracy in pendulum identification, characterized by a phase offset, suggests that parameters extracted capture the forced response, while free response is out of phase.

Additionally, the estimator introduced a 3 to 4 sample lag in velocity reconstruction.

These factors, combined with the disturbances, highlight the significant impact of hardware constraints and unaccounted non-linear dynamics on the theoretical control design.

7. Conclusions

This project successfully implemented a multi-stage control architecture for the Quanser SRV02 Rotary Pendulum system. By integrating theoretical modeling with experimental identification, a robust framework was developed to address the inherent instabilities of the system. The work achieved several key milestones:

System Identification: Physical parameters, specifically the damping coefficients and motor gains, were accurately extracted through a "matching" between simulation and experiment. This ensured the model's fidelity to the physical hardware for controlling stages.

Optimal Swing-Up: Rather than using energy-based methods, an optimal trajectory was generated allowing the system to respect actuator saturation and arm-angle constraints imposed by sensor cabling.

Stable Regulation: Linear state-feedback control via pole placement successfully stabilized the pendulum at its unstable equilibrium, with integral action added to the arm position to eliminate steady-state errors.

State Estimation: A discrete-time Kalman Filter was essential for mitigating high-frequency noise inherent in direct velocity differentiation. By tuning the diagonal process (Q) and measurement (R) noise covariance matrices, the filter provided the smooth state estimates required for stable real-time feedback.

While experimental results largely matched simulation, observed discrepancies in the later stages of free decay and stabilization highlight the influence of unmodeled dynamics, such as cable tension, frictions and actuator backlash. Ultimately, the integration of classical control, trajectory optimization, and observer estimation proved effective in achieving reliable swing-up and balance performance on the physical system.